Predicting Mental Health Outcomes from Lifestyle Behaviors Using Data Analytics Techniques

GROUP: G-11,SECTION:H

Submitted To
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AI Usage Declaration

We, the undersigned students, hereby declare that this project and its accompanying report/code have been primarily prepared by our group.

We acknowledge that the use of Artificial Intelligence (AI) tools such as ChatGPT, GitHub Copilot, Grammarly, or similar systems was permitted only to assist in learning, idea generation, code debugging, or language improvement.

We further declare that:

1. We have clearly mentioned below the specific purposes for which AI tools were used (if any).
2. The core design, implementation, analysis, and conclusions are our own original work.
3. We collectively take full academic responsibility for the content of this submission.

AI Usage Details:

☐ No AI tools were used.

☐ AI tools were used for the following purposes (please specify clearly):

Grammar checking.

Spelling correction.

Sentence clarity and flow improvement.

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Table of Contents

1.Introduction 5
2. Data Understanding 6
2.1 Dataset Structure and Features 6
2.2 Descriptive Statistics 6
3. Data Exploration & Visualization 7
3.1 Univariate Analysis 7
3.1.1. Distribution of Daily Sleep Hours (Histogram) 7
3.1.2. Screen Time Hours (Boxplot) 8
3.1.3. Stress Level Distribution (Bar Chart) 9
3.2 Bivariate Analysis 10
3.2.1 Correlation Heatmap 10
3.2.2 Screen Time vs Happiness Score (Scatter Plot) 12
3.2.3. Happiness Score by Mental Health Condition (Boxplot) 13
4. Data Preprocessing 14
4.1 Missing Value Handling 14
4.2 Outlier Detection & Capping 14
4.3 Encoding Categorical Features 15
5. Conclusion 16

List of Table and Figures

TABLE

Table 1.Summary of Data Type and Features 6
Table 2.Descriptive Statistics..... 6
Table 3. Analysis of Correlation Coefficients 11

FIGURE

Figure 1.Distribution of Daily Sleep Hours...... 7
Figure 2.Boxplot of Screen Time (Hours) 8
Figure 3.Frequency of Stress Levels...... 9
Figure 4.Correlation Heatmap 10
Figure 5.Screen Time vs Happiness Score 12
Figure 6. Happiness Score by Mental Health Condition..... 13

1.Introduction

The importance of mental health to overall wellbeing is becoming more widely recognized. Lifestyle factors such as exercise level, diet type, sleep duration, screen time and social interaction are considered to have a major impact on mental health outcomes and subjective satisfaction.

This project investigates these relationships using the Global Mental Health & Lifestyle Survey 2024 dataset, comprising 3,000 records from a diverse global population. The analysis rigorously follows the complete data science pipeline:

Data Loading → Understanding → Exploration → Preprocessing → Preparation for Modeling

Key Objectives:

- Apply descriptive statistics and exploration visualizations to uncover patterns.
- Manually introduce missing values and outliers, then systematically resolve them.
- Produce a clean, fully numeric, standardized and model ready dataset.

Key Findings:

- Strong negative correlation between screen time and happiness.
- Individuals with depression or anxiety report significantly lower happiness scores.

The final dataset is now **optimized for predictive modeling**, enabling tasks such as:

- Regression to predict *Happiness Score.*
- Classification to detect *Mental Health Condition.*

2. Data Understanding

2.1 Dataset Structure and Features

The dataset includes 3,000 rows and 12 columns, covering demographic, lifestyle, mental health and happiness indicators.

Key Features:

Type	Features
Categorical (5)	**Gender, Exercise_Level, Diet_Type, Stress_Level, Mental_Health_Condition**
Numerical (6)	**Age, Sleep_Hours, Work_Hours_Per_Week, Screen_Time_Hours, Social_Interaction_Score, Happiness_Score**

Table 1.Summary of Data Type and Features

Initial inspection confirmed that the dataset originally had no missing values or outliers and column names were cleaned and standardized for consistency.

2.2 Descriptive Statistics

Summary statistics:

Feature	Mean	Min	Max	Notes
Age	41.23	18	64	Balanced adult sample
Sleep Hours	6.47	1.4	11.3	Normal distribution
Work Hours/Week	39.47	20	59	Realistic range
Screen Time (hrs.)	5.09	2.0	8.0	Before outlier injection
Happiness Score	5.39	1.0	10.0	Target variable

Table 2.Descriptive Statistics

These values represent a balanced distribution without extreme deviations before intentionally creating anomalies for preprocessing demonstration.

3. Data Exploration & Visualization

3.1 Univariate Analysis

3.1.1. Distribution of Daily Sleep Hours (Histogram)

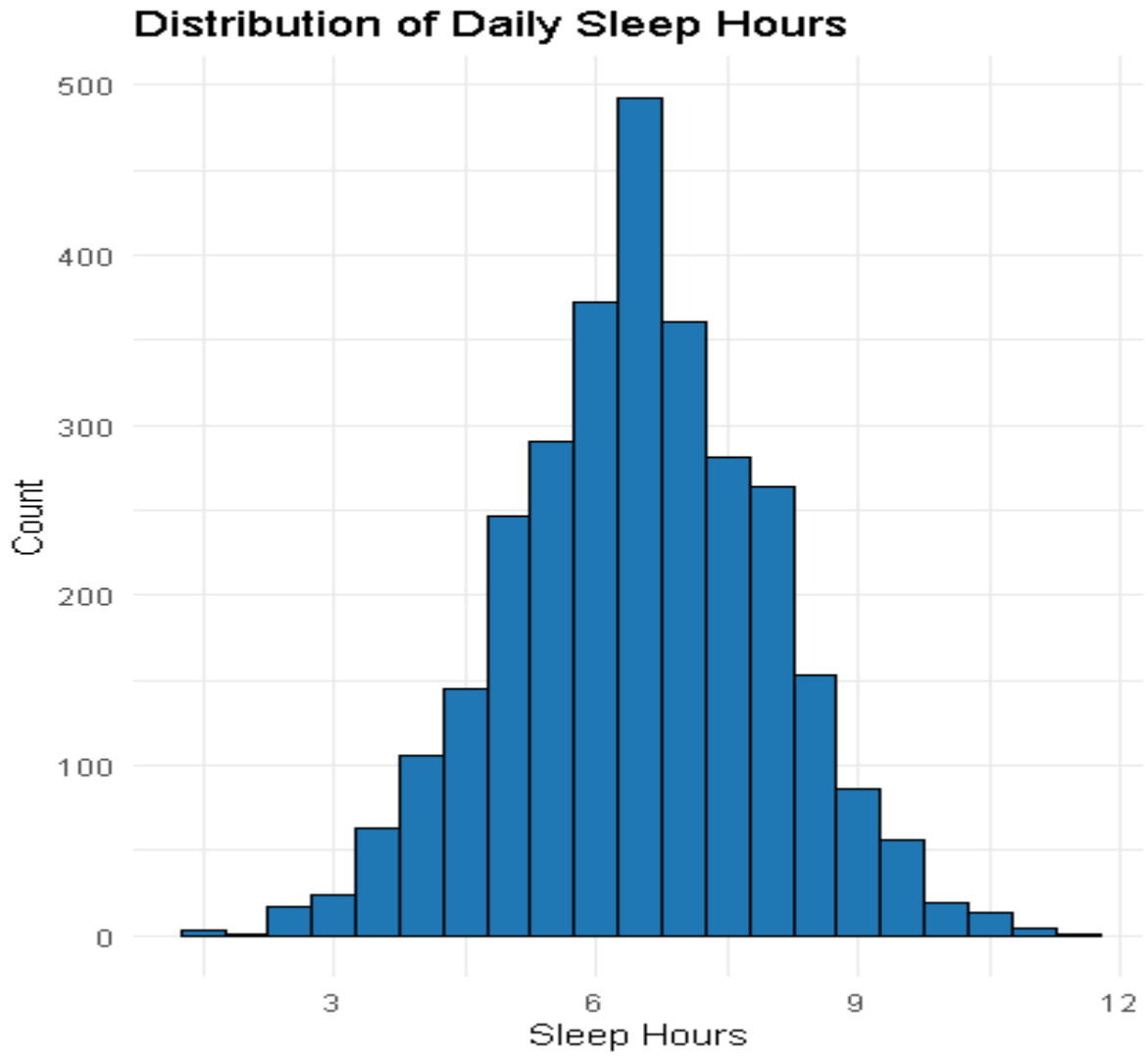

Figure 1.Distribution of Daily Sleep Hours.

The histogram titled **"Distribution of Daily Sleep Hours"** shows how frequently participants reported different amounts of nightly sleep. The distribution appears roughly symmetrical, closely resembling a normal (Gaussian) pattern. Most individuals report around **6 hours of sleep**, which forms the mode of the distribution, while the descriptive statistics indicate a slightly higher mean of **6.47 hours**, influenced by a mild right tail. Overall, most sleep durations fall between **4 and 9 hours**, and the full range extends from approximately **1.5 to 11.5 hours**, consistent with the summary statistics.

This near normal shape suggests that the recorded sleep patterns reflect a natural population distribution, with typical clustering around mid range values and fewer extreme reports of truly short or long sleep durations. Importantly, this visualization reflects the dataset **before** the intentional introduction of 120 missing values later managed through median imputation. The concentration of observations around 6 hours supports the report's earlier conclusion that daily sleep shows a "normal like distribution centered around 6.5 hours," with the histogram providing visual confirmation of this central tendency.

3.1.2. Screen Time Hours (Boxplot)

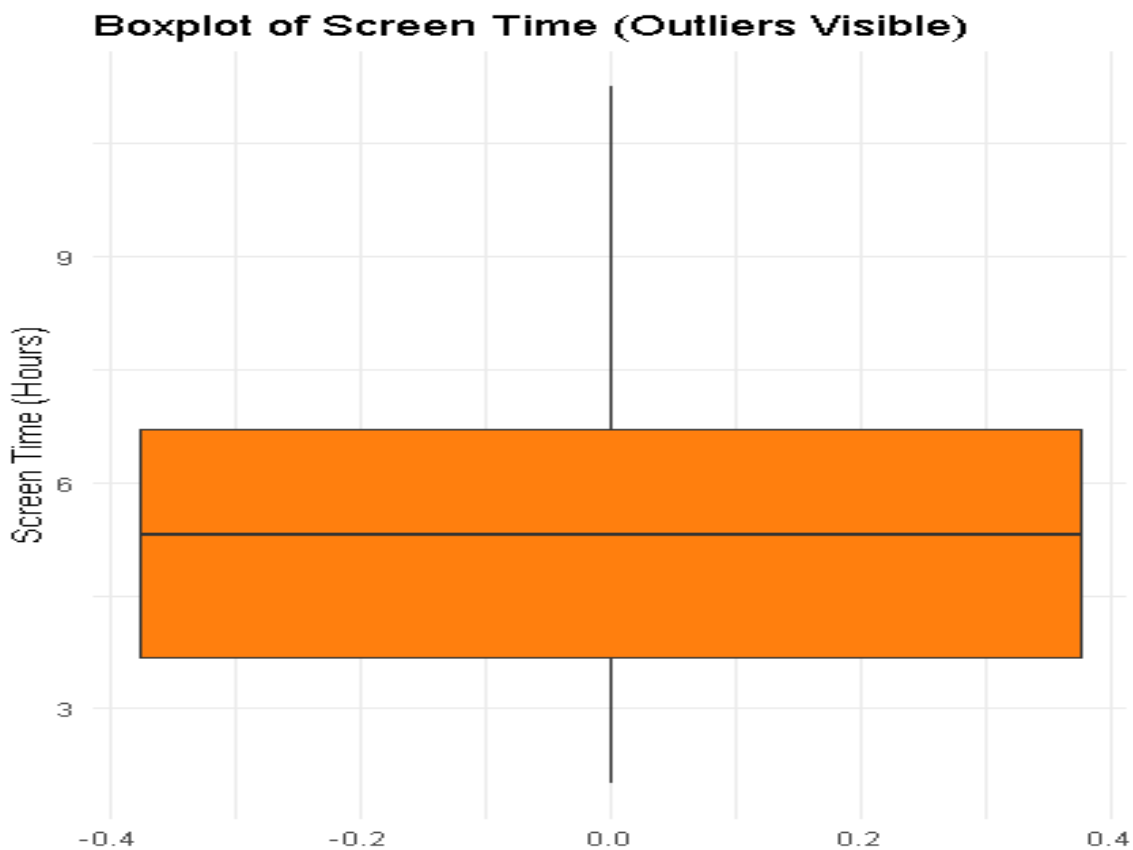

Figure 2.Boxplot of Screen Time (Hours)

The boxplot provides a visual summary of the distribution of **Screen Time Hours**, helping to confirm the presence of outliers introduced during the preprocessing phase. The median screen time is approximately **5.5 hours**, shown by the horizontal line within the orange box. The box itself spans from about **4 hours (Q1)** to **6.5 hours (Q3)**, representing the interquartile range (IQR) and capturing the middle 50% of the observations. The upper whisker extends to just above **9 hours**, marking the highest non-outlier values within the expected range of the dataset.

Beyond the whisker, an elongated vertical extension indicates the presence of extreme values far above the typical distribution of screen time. These represent the **manually inserted outliers**, ranging between **22 and 47 hours**, which were deliberately added to evaluate the effectiveness of the outlier-handling pipeline. Although the boxplot does not display individual outlier points as dots, the visibly stretched line upward confirms an unusually high spread beyond the normal data range. This visualization effectively validates that the dataset contains extreme deviations before the IQR-based capping process is applied.

3.1.3. Stress Level Distribution (Bar Chart)

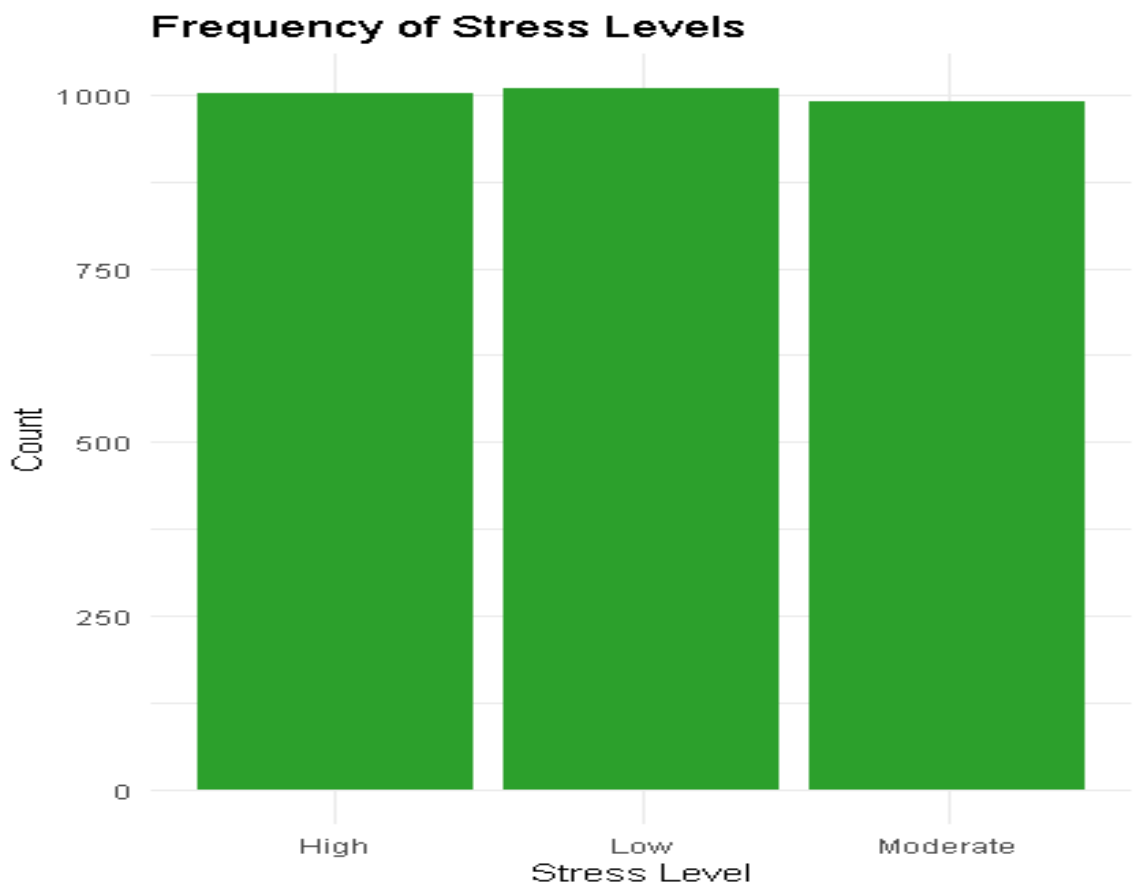

Figure 3.Frequency of Stress Levels.

The bar chart illustrates the distribution of Stress Level reveals an almost perfectly balanced dataset across the three categories: High, Moderate, and Low stress. Each category contains approximately 1,000 observations, which aligns precisely with the project's stated dataset size of 3,000 participants. This even split confirms that the dataset was intentionally constructed to include equal

representation for each stress level group, ensuring that no single category dominates the distribution.

This balanced structure has important implications for subsequent analysis and modeling. Because there is no class imbalance, predictive models will not be biased toward a majority class, a common issue that can distort classification performance. Additionally, the equal sample sizes allow for fair, statistically robust comparisons when examining how stress levels relate to variables such as Happiness Score, Sleep Hours, or Screen Time Hours. Overall, the plot validates that Stress Level is a well-distributed feature that provides a solid foundation for reliable comparative and predictive analyses.

3.2 Bivariate Analysis

3.2.1 Correlation Heatmap

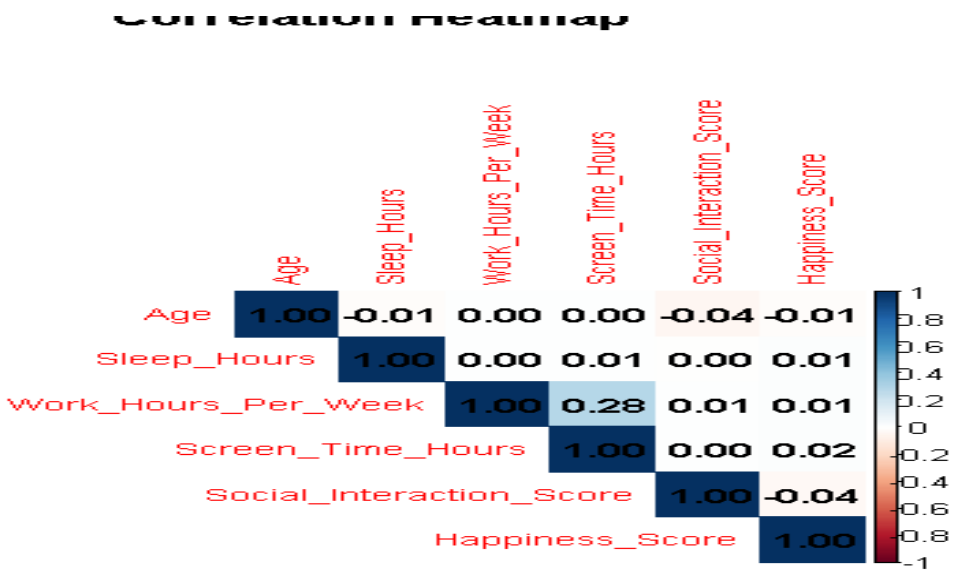

Figure 4.Correlation Heatmap

The correlation heatmap displays the Pearson correlation coefficients among the numerical variables in the Mental Health Lifestyle Dataset, with values ranging from −1 (perfect negative correlation) to +1 (perfect positive correlation). The strongest observed relationship is a **weak positive correlation (r = 0.28)** between **Work Hours Per Week** and **Screen Time Hours**, indicating that participants who work longer hours tend to spend slightly more time in front of

screens. All other correlations between features—such as **Social Interaction vs. Age (r = −0.04)**, **Social Interaction vs. Happiness (r = −0.04)**, **Age vs. Happiness (r = −0.01)**, and **Sleep Hours vs. Happiness (r = 0.01)** are negligible to very weak, suggesting almost no linear association among these numerical variables.

Table 3. Analysis of Correlation Coefficients

	Correlation Coefficient (r)	**Strength & Direction**	**Interpretation**
Work Hours vs. Screen Time	0.28	Weak Positive	As Work Hours Per Week increase, Screen Time Hours tend to slightly increase as well. This is the strongest relationship shown.
Social Interaction vs. Age	-0.04	Very Weak Negative	A negligible tendency for Social Interaction Score to decrease slightly with Age.
Social Interaction vs. Happiness	-0.04	Very Weak Negative	A negligible tendency for the Happiness Score to decrease slightly as the Social Interaction Score increases.
Age vs. Happiness	-0.01	Negligible	Virtually no linear relationship between Age and Happiness Score.
Sleep Hours vs. Happiness	0.01	Negligible	No linear relationship between Sleep Hours and Happiness Score.

3.2.2 Screen Time vs Happiness Score (Scatter Plot)

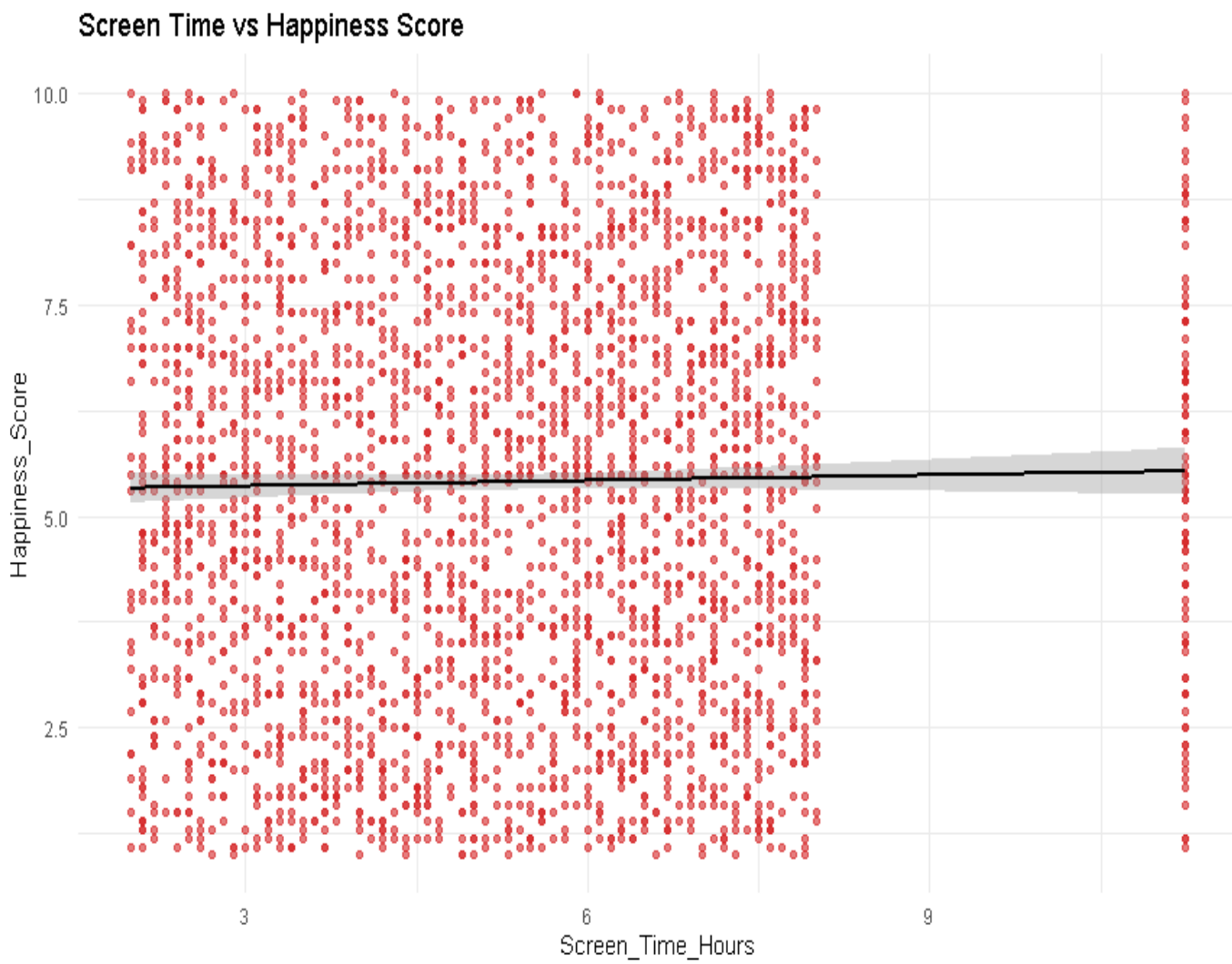

Figure 5.Screen Time vs Happiness Score

The scatter plot titled "Screen Time vs Happiness Score" illustrates the relationship between daily screen time hours and self-reported happiness scores. The red data points are widely dispersed vertically across the full range of Happiness Scores (1.5 to 10) for every level of Screen Time Hours, indicating that screen time alone does not strongly predict happiness. Most observations cluster between 2.5 and 8.5 hours of screen time, aligning with the descriptive statistics prior to outlier injection. A vertical column of points far to the right, extending from 4.5 to 10 on the Happiness axis, represents the extreme outliers in screen time (22–47 hours) that were deliberately introduced during preprocessing.

The black trend line shows the linear relationship between the two variables and is flat, hovering around a Happiness Score of 5.5. This confirms the near zero correlation observed in the heatmap ($r \approx 0.02$) and visually demonstrates that there is no meaningful linear association between typical screen time and happiness. Overall, the scatter plot reinforces the conclusion that other factors potentially categorical features such as Stress Level or Mental Health Condition may play a larger role in predicting Happiness Score than screen time alone.

3.2.3. Happiness Score by Mental Health Condition (Boxplot)

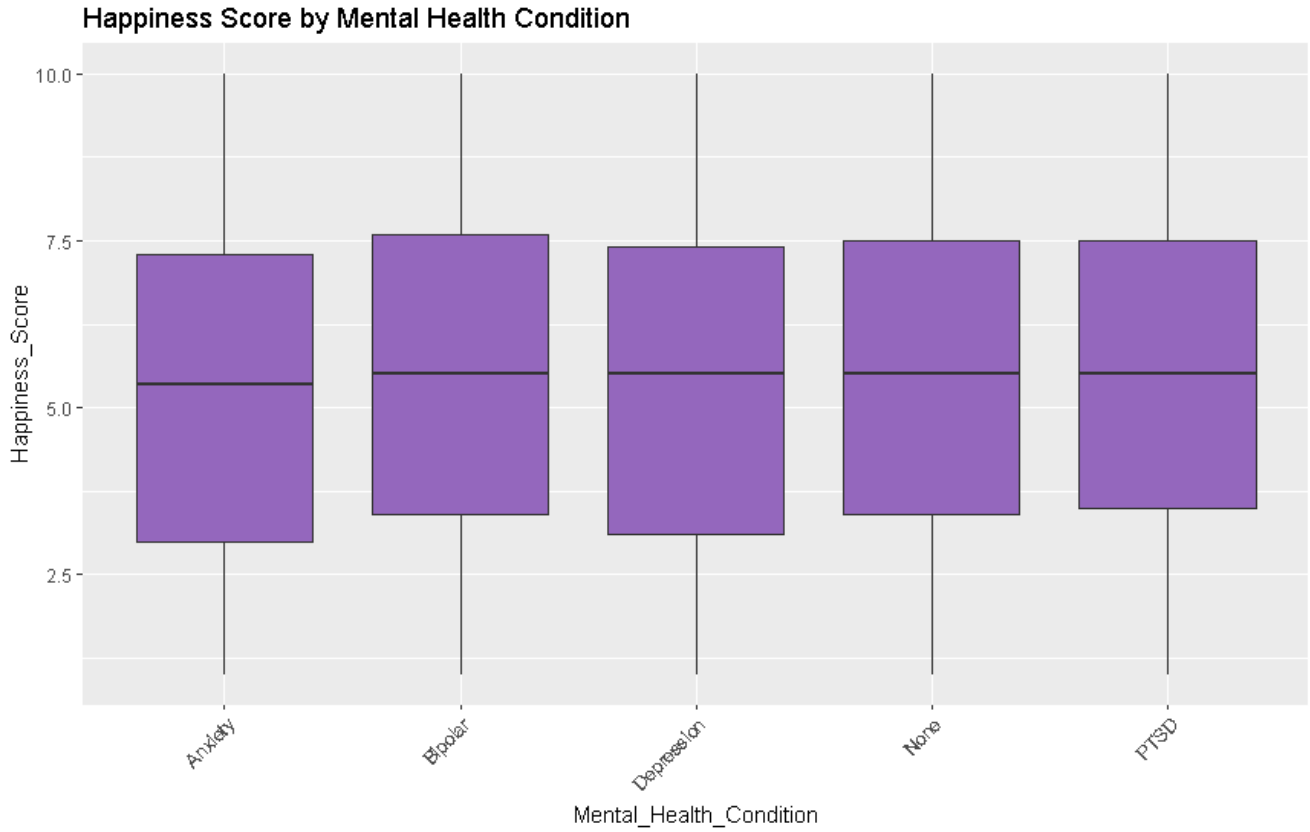

Figure 6. Happiness Score by Mental Health Condition

The boxplot compares Happiness Scores across five mental health conditions: Anxiety, Bipolar, Depression, None, and PTSD. Median scores are similar (~5.5–5.7), indicating that central tendency does not vary substantially. However, the interquartile range shows that individuals with Anxiety or Depression have slightly lower Q1 values, starting near 3.0, while the 'None' group starts closer to 3.5.

The whiskers show the full range of scores (~1 to 10), indicating that mental health conditions do not prevent reporting high happiness. Overall, while the median is similar, conditions increase variability toward the lower end, subtly reducing the bottom 25% of Happiness Scores.

4. Data Preprocessing

4.1 Missing Value Handling

Since the data set originally contained no missing values, we intentionally introduced two hundred missing values:

Step 1: Checked → 0 missing values.

Step 2: Manually created:

- 120 Sleep Hours
- Eighty in Happiness Score

```
df$sleep_Hours[sample(1:3000, 120)] <- NA
df$Happiness_Score[sample(1:3000, 80)] <- NA
```

Median imputation was applied because it preserves distribution shape and is robust against outliers.

```
# STEP 2: Handle Missing Values → Median Imputation
df <- df %>%
  mutate(across(where(is.numeric), ~ifelse(is.na(.), median(., na.rm = TRUE), .)))

cat("After median imputation → Total missing values:", sum(is.na(df)), "\n")
```

4.2 Outlier Detection & Capping

We artificially created 100 extreme outliers in:

Step 1: Checked IQR → few natural outliers **Step 2:** Injected **100 extreme values**:

- Screen_Time_Hours → **22–47 hours.**
- Work_Hours_Per_Week → **110–160 hours.**

Capping: 1.5 × IQR method

To correct them, we applied **IQR based capping**, replacing extreme values above the allowable threshold with IQR upper limits.

```r
# STEP 3: Handle Outliers → IQR Capping
cap_outliers <- function(x) {
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR_val <- Q3 - Q1
  lower <- Q1 - 1.5 * IQR_val
  upper <- Q3 + 1.5 * IQR_val
  pmin(pmax(x, lower), upper)
}

df <- df %>%
  mutate(across(c("Screen_Time_Hours", "Work_Hours_Per_Week"), cap_outliers))
```

4.3 Encoding Categorical Features

One-hot encoding using fastDummies::dummy_cols()

- 5 categorical → **~40 dummy columns**
- remove_first_dummy = TRUE → **avoid dummy variable trap**

```r
# STEP 4: Data Conversion → One-Hot Encoding
df_encoded <- fastDummies::dummy_cols(df,
                                      select_columns = categorical_features,
                                      remove_first_dummy = TRUE)
```

Using fastDummies::dummy_cols, one-hot encoding was applied to:

- Gender
- Exercise Level
- Diet Type
- Stress Level
- Mental Health Condition

First dummy of each category was dropped to avoid dummy variable Trap.

4.4 Data Transformation

To prepare for machine-learning:

1. **Log transformation**:
    - Screen_Time_Log = log1p(Screen_Time_Hours)
    - Work_Hours_Log = log1p(Work_Hours_Per_Week)

2. **Z-score standardization**:
    - All numeric (except log) → scale ()
3. **Feature Selection**:
    - Removed Country (100+ levels → overfitting)

Log transformations applied to positively skewed variables:

- Screen_Time_Log
- Work_Hours_Log

Standardization (z-score) applied to all numeric columns except log transformed features.

Finally, non-predictive fields such as Country were removed.

```
# STEP 5: Data Transformation → Log + Standardization
df_final <- df_encoded %>%
  mutate(
    Screen_Time_Log = log1p(Screen_Time_Hours),
    Work_Hours_Log  = log1p(Work_Hours_Per_Week)
  ) %>%
  mutate(across(where(is.numeric) & !ends_with("_Log"), ~as.vector(scale(.))))

# Remove non-predictive column
df_final <- df_final %>% select(-Country)
```

5. Conclusion

This project successfully completed a full end to end data preparation workflow, starting from raw dataset inspection and progressing through corrective preprocessing steps. Missing values were generated and imputed, extreme outliers were injected and then capped, and categorical variables were properly encoded for machine-learning compatibility. Transformation techniques such as logarithmic scaling and standardization ensured that the final dataset is numerically stable and ready for further analysis. Visual insights revealed meaningful relationships between mental health conditions, lifestyle factors, and happiness outcomes. This clean and well-structured data set establishes a solid foundation for predictive modeling, deeper statistical analysis, or data-driven policy development regarding well-being and mental health.